

XJTLU Entrepreneur College (Taicang) Cover Sheet

Module code and Title	DTS406TC Natural Language Processing		
School Title	School of Artificial Intelligence and Advanced Computing		
Assignment Title	Coursework 1 (CW 1)		
Submission Deadline	5 pm China time (UTC+8 Beijing) on 22/May/2026		
Final Word Count	3000		
If you agree to let the university use your work anonymously for teaching and learning purposes, please type “yes” here.			

I certify that I have read and understood the University’s Policy for dealing with Plagiarism, Collusion and the Fabrication of Data (available on Learning Mall Online). With reference to this policy I certify that:

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Scoring – For Tutor Use	
Student ID	

Stage Marking of	Marker Code	Learning Outcomes Achieved (F/P/M/D) (please modify as appropriate)			Final Score
		A	B	C	
1 st Marker – red pen					
Moderation – green pen	IM Initials	The original mark has been accepted by the moderator (please circle as appropriate):			Y / N
		Data entry and score calculation have been			Y

		checked by another tutor (please circle):			
2 nd Marker if needed – green pen					
For Academic Office Use			Possible Academic Infringement (please tick as appropriate)		
Date Received	Days late	Late Penalty	☐ Category A	Total Academic Infringement Penalty (A,B, C, D, E, Please modify where necessary) _____	
			☐ Category B		
			☐ Category C		
			☐ Category D		
			☐ Category E		

Policy

The report should be submitted in PDF format. The word count limit of the report is 3,000. All code must be written in Python. It should be well-structured, easy to read, and thoroughly documented. Each function should include comments explaining its purpose and functionality. Ensure that the code runs without errors. All required files should be included in a single zip file. A README file is needed to explain how to run the code and list any dependencies. The document must be submitted through Learning Mall Online to the appropriate drop box.

Importance

AI is permitted as a learning aid for your coursework, but it must not be used to auto-generate entire sections of code or the final report.

DTS406TC Natural Language Processing

Coursework 1 (Group Assessment)

Due: **5:00 pm China time** (UTC+8 Beijing) on **22/May/2026**

Weight: **40%**

Maximum score: 100 marks (80 % group marks + 20 % individual marks)

Groupings: **Each group consists of 2 or 3 students. Try to form groups of 3 students. You are free to select your own team members. Students who do not make a selection will be randomly assigned to a team. Once the teams are confirmed, no changes will be permitted.**

Assessed learning outcomes:

A Systematically comprehend the theoretical foundations of Natural Language Processing

B Apply statistical and machine learning techniques to process and analyze natural language data

Overview

Document topic classification is a core NLP task that automatically assigns predefined topic labels to documents based on their textual content. It is widely used in applications such as news categorization, academic paper indexing, and social media content organization. Recent advances in deep learning and large language models have significantly improved topic classification performance by capturing richer contextual and semantic representations of documents.

1. Literature Review on Document Topic Classification (20 Marks, Individual Work)

- a) Provide an overview of document topic classification and its applications. Please include three examples of real-life applications of document topic classification. (6 Marks)
- b) List and explain three key challenges in document topic classification. (6 Marks)
- c) Elaborate on two traditional methods (e.g., Naïve Bayes, SVM) and two deep learning approaches (e.g., LSTM, GRU) for document topic classification. Discuss the advantages and disadvantages of each approach. (8 Marks)

**** Each team member must complete the literature review independently. This section will be assessed on an individual basis.**

2. Data Preprocessing (12 Marks, Group Work)

- a) Collect two datasets from platforms such as Twitter or News websites, focusing on document topic classification in different scenarios. Each dataset must contain a minimum of 3,000 instances. The two datasets should represent different classification scenarios (e.g., education discussions vs. entertainment discussions, technology forums vs. health forums). (2 Marks per dataset * 2 = 4 Marks)

****Example Datasets**

- *20 Newsgroups Dataset*: It contains approximately 20,000 documents collected from 20 different newsgroups, covering topics such as computer science and sports.

<https://archive.ics.uci.edu/ml/datasets/Twenty+Newsgroups>

- *Reuters-21578 Dataset*: It contains 21,578 news articles from the Reuters newswire service.

<https://archive.ics.uci.edu/dataset/137/reuters%2B21578%2Btext%2Bcategorization%2Bcollection>

- b) Preprocess both datasets by performing tasks such as tokenization, stopwords removal, lowercasing, and optional stemming or lemmatization. (2 Marks per dataset * 2 = 4 Marks)
- c) Conduct a statistical analysis including word frequency distribution, vocabulary size, average document length, and label distribution. (2 Marks per dataset * 2 = 4 Marks)

3. Algorithm Description & Implementation (46 Marks, Group Work)

a) Algorithm Design (16 Marks)

Select four approaches for the document topic classification task: two traditional methods and two deep learning methods. All four approaches must be applied to both crawled datasets. Provide pseudo-code for each algorithm and briefly explain the function of each component in the pseudo-code. (4 Marks per algorithm * 4 = 16 Marks)

b) System Implementation (30 Marks)

Develop a document topic classification system in Python for each approach. The implementation pipeline should include:

- 1) Feature Engineering: Convert textual data into appropriate representations (e.g., TF-IDF, word embeddings, contextual embeddings). (5 Marks)
- 2) Algorithm Implementation: Implement all four algorithms. (5 Marks per algorithm * 4 = 20 Marks)
- 3) Evaluation Metrics: Compute relevant evaluation metrics to assess model performance. (5 Marks)

****Assessment Criteria for Each Implementation (Graded on a 0–5 Scale)**

5 Marks – Outstanding: The function is fully and correctly designed and implemented. Each step is clearly defined, and all decision or judgment conditions are explicitly and logically specified.

4 Marks – Very Good: The function design and implementation are largely complete and correct. The steps are clearly described, and the judgment conditions are well defined with minor omissions.

3 Marks – Satisfactory: The function design and implementation are mostly complete. The main steps are understandable, but some details or judgment conditions lack clarity or precision.

2 Marks – Limited: The function design and implementation show partial completion. The main steps can be identified; however, the logic, structure, or judgment conditions are unclear.

1 Mark – Poor: The function is minimally implemented or poorly described, with major logical gaps or missing key components.

0 Marks – No Submission or Missing Section: This sub-task is not submitted or entirely missing.

4. Results Analysis (15 Marks, Group Work)

Present the document topic classification results for each approach applied to both datasets.

Select and apply three relevant evaluation metrics (e.g., precision, recall, and F1-score) to assess model performance. Explain the reasons behind the chosen four algorithms' performances based on each selected metric. (5 Marks per metric * 3 = 15 Marks)

****Assessment Criteria for Each Metric (Graded on a 0–5 Scale)**

5 – Outstanding: Results are comprehensively presented with clear tables/figures. Metrics are correctly computed and interpreted. Performance differences are logically analyzed with strong justification.

4 – Very Good: Results are clearly presented and mostly well explained. Metrics are correctly applied with minor gaps in interpretation.

3 – Satisfactory: Results are presented but explanations lack depth or detailed reasoning.

2 – Limited: Results are partially reported with unclear or weak explanation of metric values.

1 – Poor: Results are minimally presented or incorrectly interpreted.

0 – No Submission: Results section missing.

5. Report Writing (7 Marks, Group Work)

This coursework evaluates your understanding the challenges of the problem and the correctness of the proposed algorithms. It also tests your professional skills in terminology usage, presentation of algorithms and experimental results, as well as the logical manner of the proposal. (7 Marks)

****In detail, marks will be awarded based on:**

- Demonstration of professional knowledge
- Clarity of algorithm design
- Quality of result presentation
- Logical structure and academic writing
- Proper formatting
- Adherence to the word count range
- Overall academic standard

Submission

One of the team members must submit a single zip file. The zip file is named "TeamID_Coursework.zip". It includes: a cover letter with the group member information and the final PDF reports. The final PDF reports include the individual reports of literature review on the document topic classification and the group report. A folder labeled "algorithms" contains all the model implementations, data preprocessing scripts, and evaluation scripts. A folder labeled "data" contains all the datasets and the experimental results in the CSV format.

End of Coursework